

Evaluating Explainable AI

From Quality Properties to Metrics

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A provocation

“SHAP is better than LIME.”

Better according to what?

Faithfulness? Stability? Compactness? Human usefulness?

Predictive performance is not explanation quality

Does f predict correctly? $\neq$ Does E describe f correctly?

- Predictive accuracy compares model outputs with target labels.
- Faithfulness concerns the explanation relative to model behavior.

What is being evaluated?

$$x \xrightarrow{f} f(x) \xrightarrow{E} e \quad Q(e, f, x, D, \theta_Q)$$

x	input instance being explained
D	data or background sample
f	predictive model
$f(x)$	model output for x

E	explanation method
e	explanation produced by E
Q	evaluation procedure
θ_Q	metric/protocol parameters

We evaluate an explanation under a protocol, not in isolation.

A good explanation is not one thing

- Faithful but difficult to understand.
- Plausible but unfaithful to the model.
- Compact but incomplete.
- Stable but irrelevant to the user's task.

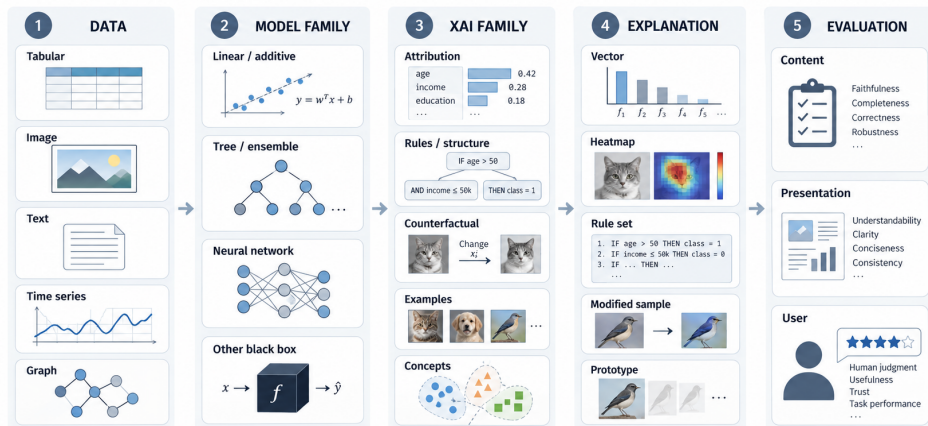
The scientific claim determines which trade-offs matter.

Data modality changes the evaluation operation

- **Tabular**: replace a feature or feature group.
- **Image**: mask a region or superpixel.
- **Text**: remove or replace a token.
- **Time series**: perturb an interval.
- **Graph**: remove a node or edge.

The same property can require different operational metrics for different representations.

A schematic XAI landscape



Content



Presentation



User

Co-12 instantiates the evaluation branch: what the explanation claims, how it is presented, and who uses it.

Three axes of an evaluation

Evaluation perspective

Where evidence comes from

Functionally grounded
Human-grounded
Application-grounded

Study design

How evidence is collected

Perturbation test
User simulation task
Expert decision study

Quality perspective

Which claim is tested

Content
Presentation
User

Example: a human-grounded simulation task can test a Content claim about faithfulness.

Keep these choices explicit: evidence source, procedure, and quality claim.

The literature uses overlapping language

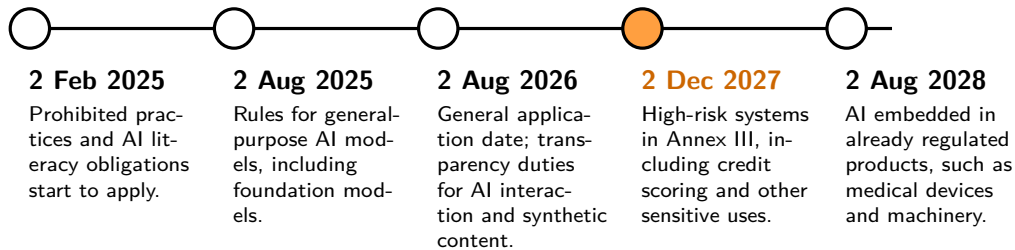
Explainability, interpretability, transparency, faithfulness, stability, robustness, comprehensibility, and actionability are related but not interchangeable.

Regulatory vocabulary: EU AI Act

The AI Act frequently uses **transparency**, **information to deployers**, **interpretation of outputs**, and human oversight language. In Recital 27, transparency is linked to traceability and explainability; this is related to XAI, but not the same as a metric definition.

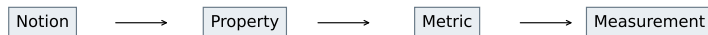
A shared vocabulary makes claims testable.

EU AI Act: application timeline



Sources: Regulation (EU) 2024/1689; Digital Omnibus on AI, Regulation (EU) 2026/1744, in force since 27 July 2026.

Four levels



Level	Meaning	Examples
Notion	Broad idea	transparency, explainability, interpretability
Property	Desired quality	faithfulness, stability, compactness, coherence
Metric	Operational test	deletion AUC, AOPC, max-sensitivity, entropy
Measurement	Protocol-bound value	deletion AUC = 0.31 with median baseline, seed 42

From property to measurement

Claim

The explanation is faithful to the predictive model.

One operationalization

Deletion AUC; replace top-ranked features by the training median.

Measurement

0.31 on a named model, dataset, target, baseline, and seed.

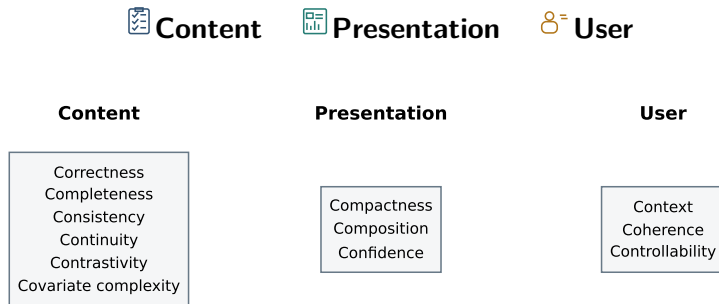
A scalar without its protocol is incomplete evidence.

The evaluation chain

Goal $\rightarrow$ Claim $\rightarrow$ Property $\rightarrow$ Metric $\rightarrow$ Protocol $\rightarrow$ Evidence

The metric is selected after the claim and property, not before them.

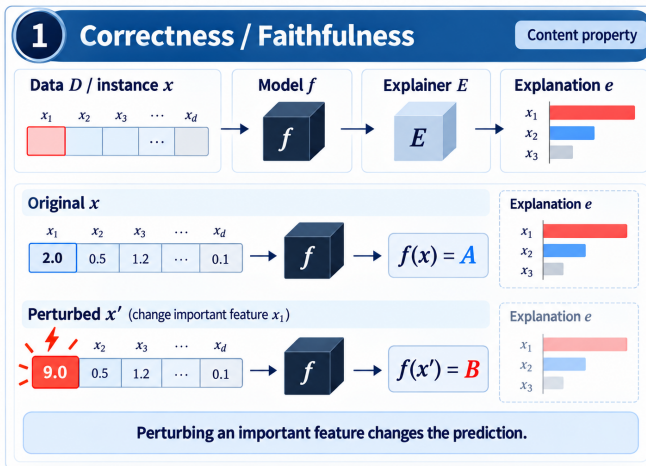
Co-12: a quality map, not a scorecard



Co-12 is a concrete quality map for the evaluation branch of the XAI landscape.[3]

Co-12: questions to ask

 Content	 Presentation	 User
Correctness / faithfulness	Compactness: concise enough?	Context: useful for this task?
Completeness	Composition: appropriate form?	Coherence: aligns with domain knowledge?
Consistency	Confidence: uncertainty represented?	Controllability: user can interact?
Continuity		
Contrastivity		
Covariate complexity		

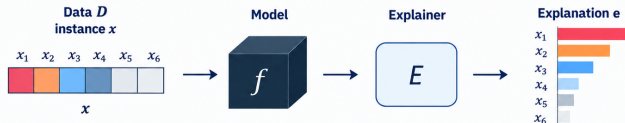


2

Completeness

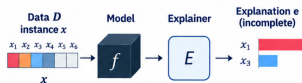
Content property

The explanation should **cover the set of relevant factors** from x that are sufficient to **justify the decision** of f .



Incomplete explanation

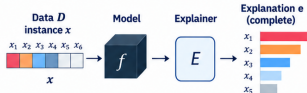
Only a subset of relevant features is shown.



The explanation omits important features (e.g., x_2 and x_4), so it is not sufficient to justify $f(x)$.

Complete explanation

It includes the relevant set of features that support the decision.

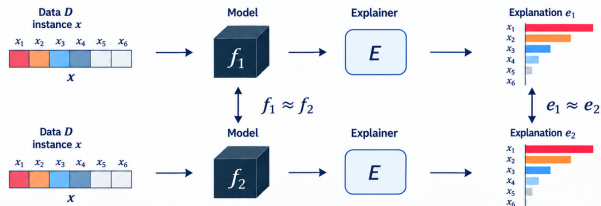


The explanation covers the relevant features (e.g., x_1, x_2, x_3, x_4), providing sufficient evidence for the decision $f(x)$.

3

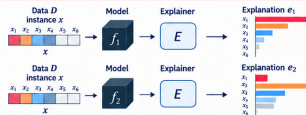
Consistency

Content property

Functionally equivalent inputs or models should produce **equivalent explanations**.

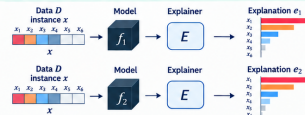
Inconsistent explanations

Equivalent models (or runs) produce different explanations.

Although $f_1 \approx f_2$ and $f(x)$ is the same, the explanations are different, which violates consistency.

Consistent explanations

Equivalent models (or runs) produce similar explanations.



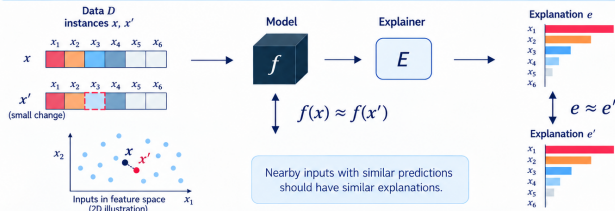
The explanations are similar, reflecting the same decision logic, which satisfies consistency.

4

Continuity

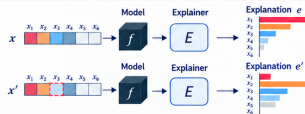
Content property

Small changes in x that do not substantially change $f(x)$ should lead to **similar explanations**.



Discontinuous explanations

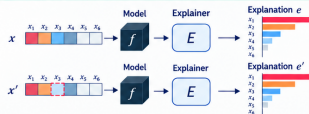
A small input change causes a large explanation change.



Although x and x' are very similar and $f(x) \approx f(x')$, the explanations differ too much, which violates continuity.

Continuous explanations

Nearby inputs produce similar explanations.

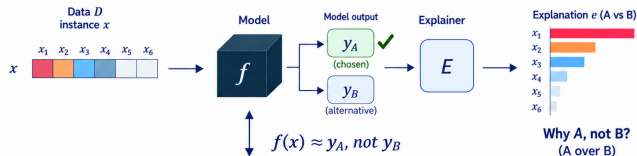


The explanations remain similar for nearby inputs, reflecting local smoothness around x .

5 Contrastivity

Content property

The explanation should clarify **why** model f chooses **one output** instead of **another** possible output.



The explanation should indicate which features favor the chosen output over the alternative output.

Non-contrastive explanation

It explains the decision, but not relative to an alternative output.



It does not explain **why** y_A was chosen instead of y_B , so the explanation is **not** contrastive.

Contrastive explanation

It identifies the features that distinguish the chosen output from the alternative.



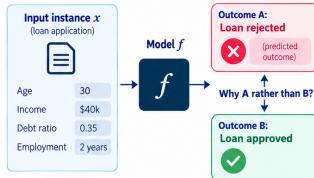
The explanation highlights the features that favor y_A over y_B , which satisfies contrastivity.

Contrastive vs Counterfactual Explanations

Two related but different ways of explaining model decisions.

Contrastive explanation

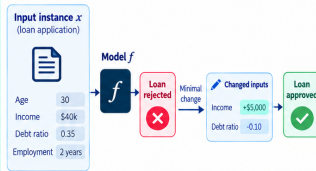
Compares the chosen outcome with an alternative foil, without necessarily changing the input.



The explanation answers why the model chose the predicted outcome (A) instead of an alternative outcome (B) for **the same input**, highlighting the factors that distinguish them.

Counterfactual explanation

Changes the input as little as possible until the prediction flips.



The explanation identifies the **smallest changes to the input features** that would result in a different outcome, showing a plausible alternative scenario.

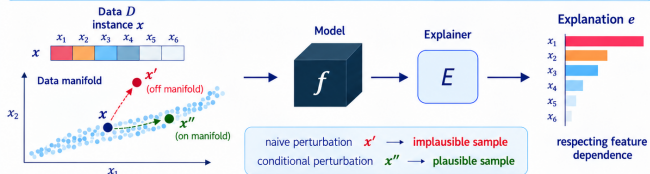


Contrastive explains the difference between outcomes; counterfactual shows what changes would flip the outcome.

6 Covariate complexity

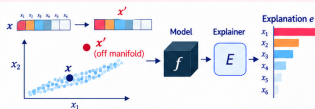
Content property

The explanation and its evaluation should **respect dependencies among features** in data D , **avoiding unrealistic perturbations** outside the data manifold.



Naive perturbation

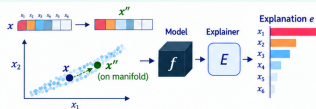
Independent feature changes can break covariate structure.



Perturbing one feature in isolation can create unrealistic samples, so the explanation may not reflect behavior on plausible data.

Covariate-aware perturbation

Perturbations should preserve plausible feature relationships.

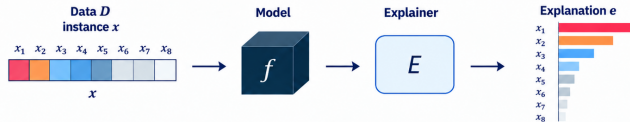


Conditional or structured perturbations stay on the manifold, so the explanation better respects the statistical structure of D .

7 Compactness

Presentation property

The explanation should be **small and concise** enough to be easily inspected and understood.



Non-compact explanation

Too many features are included, including marginal ones.

Explanation e (non-compact)

x_1
 x_2
 x_3
 x_4
 x_5
 x_6
 x_7
 x_8

Long
explanation
with many small
contributions.

Example explanation (text)

IF $x_1 > 0.5$ (important)
AND $x_2 \leq 1.2$ (important)
AND $x_3 > 0.1$ (moderate)
AND $x_4 \leq 0.3$ (small)
AND $x_5 > 0.0$ (small)
AND $x_6 \leq 0.8$ (small)
AND $x_7 > -0.2$ (very small)
AND $x_8 \leq 1.5$ (very small)
THEN $f(x) = \text{class A}$



The explanation is not compact, because it uses many factors, including marginal ones, making it harder to inspect.

Compact explanation

Only the key factors are kept.

Explanation e (compact)

x_1
 x_2
 x_3

Small set of
relevant factors
used to explain
the decision.

Example explanation (text)

IF $x_1 > 0.5$ (important)
AND $x_2 \leq 1.2$ (important)
AND $x_3 > 0.1$ (moderate)
THEN $f(x) = \text{class A}$



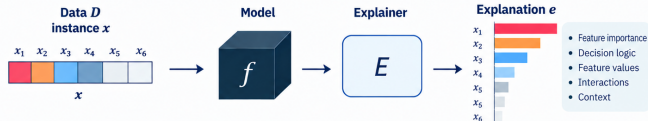
The explanation is compact, using a small and concise set of relevant factors, which improves understandability and usability.

8

Composition

Content property

The explanation should combine **multiple types of information** to provide a more complete understanding.

**Poor composition**

The explanation includes only a single type of information (e.g., feature importance).

Explanation e 

Only feature importance

Missing complementary information

- ✗ **Decision logic**
(e.g., predicates, rules)
- ✗ **Feature values**
(e.g., $x_1 = 0.8$)
- ✗ **Feature interactions**
(e.g., x_1 and x_3)
- ✗ **Context information**
(e.g., prediction score, class)



The explanation is not well composed, as it provides only one type of information, which limits understanding.

Well-composed explanation

The explanation combines multiple types of information to better support the decision.

Explanation e (composed)**Complementary information included**

- Decision logic**
(e.g., IF $x_1 > 0.5$ AND $x_3 \leq 1.2$)
- Feature values**
(e.g., $x_1 = 0.8$, $x_3 = 0.4$)
- Feature interactions**
(e.g., x_1 and x_3 together increase the prediction)
- Context information**
(e.g., prediction score = 0.76, class = A)



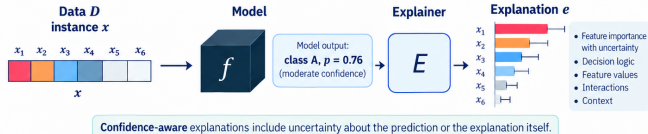
The explanation is well composed, combining feature importance, decision logic, feature values, interactions, and context information, providing a more complete and understandable explanation.

9

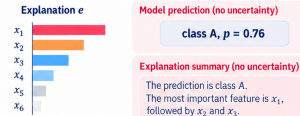
Confidence

Presentation property

The explanation should **communicate uncertainty**, showing how certain the model or explanation is about the decision.

**Overconfident explanation**

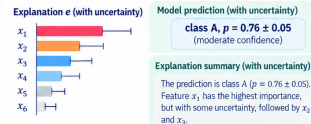
It presents the explanation as certain, without uncertainty information.



The explanation hides uncertainty, which can make the evidence look more certain than it really is.

Confidence-aware explanation

It shows uncertainty about the prediction or the explanation.

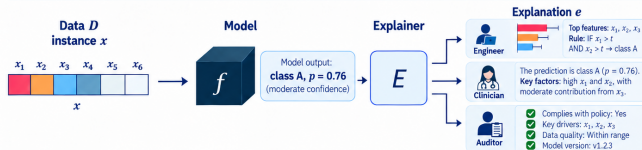


The explanation communicates uncertainty, helping the user judge how reliable the explanation and decision are.

10 Context

User property

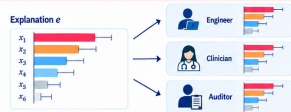
The explanation should **fit the user, task, and decision context**, providing the information that is **most relevant** for its intended use.



The same model decision may require **different explanations** for **different users and goals**.

Context-insensitive explanation

The same explanation is shown to every user.



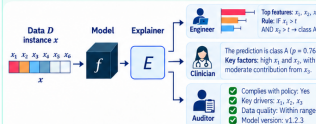
A one-size-fits-all explanation may omit the information each user actually needs.



The explanation ignores user needs and task context, so it may be less useful in practice.

Context-aware explanation

Different users receive explanations suited to their needs.



The explanation is adapted to the user's role, decision task, and downstream use.



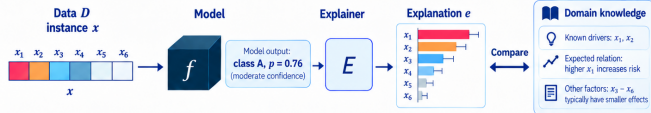
The explanation is context-aware, making it more relevant, actionable, and useful for each stakeholder.

11

Coherence

User property

The explanation should be **compatible with relevant domain knowledge** and prior expectations, while noting that plausibility alone does not prove faithfulness.



Coherence compares the explanation with what experts or science expect about the domain. Coherence supports usability and trust, but it should be evaluated separately from correctness.

Incoherent explanation

It conflicts with relevant domain knowledge.



The explanation assigns highest importance to x_6 , which is not a known key driver according to domain knowledge.



The explanation highlights factors that contradict domain knowledge, so users may find it implausible or suspicious.

Coherent explanation

It aligns with relevant domain knowledge.



The explanation highlights x_1 and x_2 as the main factors, which agree with domain knowledge and expected patterns.



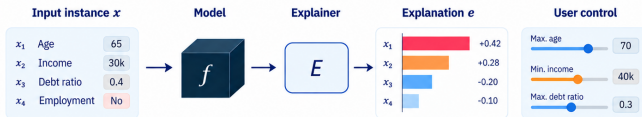
The explanation is coherent because its main factors agree with relevant expert knowledge and expected patterns.

12

Controllability

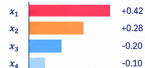
User property

The explanation should allow users to **influence and steer the decision process** by adjusting relevant factors or constraints in a transparent and predictable way.



Non-controllable explanation

The explanation only describes the decision, without enabling user control or what-if analysis.

Explanation e (static)

Limitation

The user can see which factors influenced the decision, but cannot interact with the explanation or explore alternative scenarios.



The explanation is not controllable, as it does not allow the user to modify inputs, constraints or preferences to analyze their impact on the decision.

Controllable explanation

The explanation allows users to explore how changes in relevant factors affect the decision, supporting what-if analysis and control.

Interactive explanation



User adjustment (what-if)



Updated prediction

$f(x) =$
Approve
 (previously: Reject)



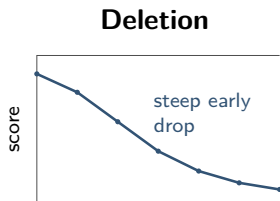
The explanation is controllable, allowing the user to modify relevant factors and observe their impact on the decision, which improves transparency, trust and usability.

One content property, multiple metrics

Level	Content
Perspective	 Content explanation claim about the model/data relationship
Property	Correctness / faithfulness : does $E(f, x)$ track the behavior of f around x ?
Metric families	Perturbation response : deletion, insertion, AOPC Attribution agreement : faithfulness correlation, infidelity Sanity controls : randomization, synthetic / white-box checks
Protocol choices	Perturbation : baseline, masking rule, feature grouping Measurement : normalization, number of samples, random seed

Same property, different operationalizations: each metric exposes different assumptions and failure modes.[2]

Deletion, insertion, and AOPC



features removed



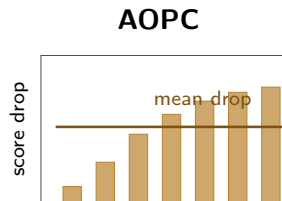
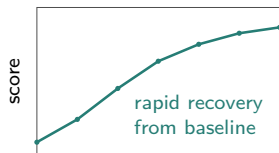
start full

mask top features

features inserted



start baseline reveal top features



perturbation step k

$$\frac{1}{K} \sum_k [f(x) - f(x^{\text{del}(k)})]$$

All three depend on the same choices: ranking, baseline, replacement rule, feature grouping, and target score.

Properties can conflict



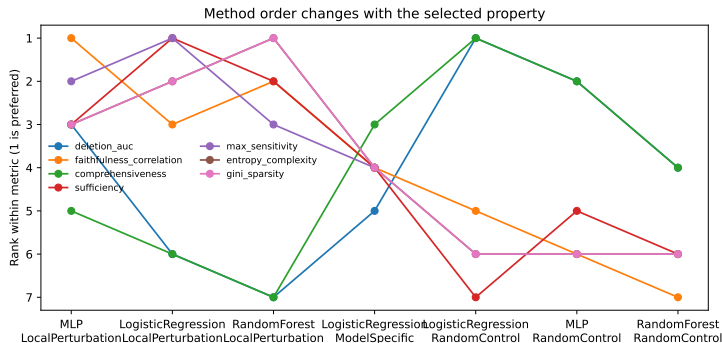
Content: a faithful explanation may be verbose.



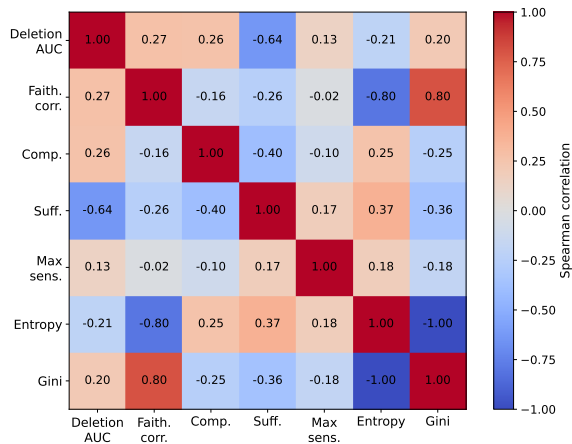
Presentation: a compact explanation may omit relevant behavior.

The evaluation objective must state which trade-offs are acceptable.

Metric disagreement



Metrics do not necessarily agree



How to read

- Each cell compares two metric outputs.
- Strong agreement means similar rankings, not identical meaning.
- Weak or negative agreement flags a trade-off or protocol effect.

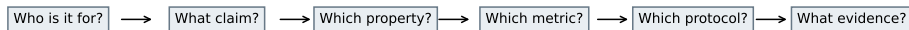
This is benchmark-specific evidence, not a universal ordering.

Discussion

- Better deletion AUC but worse stability: which method is better?
- If deletion is out of distribution, what did the score test?
- What control should a benchmark be able to reject?

Answer by naming the claim and intended use.

A defensible evaluation workflow



Then specify baselines, controls, aggregation, uncertainty, and the need for human evaluation.

Take-home points

1. Explanation quality is multidimensional.
2. A property and a metric are not the same thing.
3. Metrics encode assumptions about perturbations, baselines, and outputs.
4. Technical validity and human usefulness are different questions.
5. Strong evaluation combines complementary evidence and controls.

Final question

Do not evaluate an explanation.

Evaluate claims about an explanation.

Which claim? Which property? Which metric? Which protocol? Compared with what control?

References I

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Sanity checks for saliency maps.
In Advances in Neural Information Processing Systems, 2018.
-  A. Hedstr"om, L. Weber, D. Krakowczyk, D. Bareeva, F. Motzkus, W. Samek, S. Lapuschkin, and M. M.-C. H"ohne.
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Journal of Open Source Software, 2023.

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-  M. Nauta, J. Trienes, S. Pathak, E. Nguyen, C. Peters, Y. Schmitt, J. Schl"otterer, M. Van Keulen, and C. Seifert.
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ACM Computing Surveys, 2023.
-  C.-K. Yeh, C.-Y. Hsieh, A. Suggala, D. I. Inouye, and P. Ravikumar.
On the (in)fidelity and sensitivity of explanations.
In *Advances in Neural Information Processing Systems*, 2019.

Faithfulness: perturb important features

Perspective:  **Content**

Property: Correctness /
faithfulness

Question: Do important attributed features
control the model output?

Let $a = E(f, x)$, where a_i is feature importance. A perturbation-based test asks whether replacing features with large $|a_i|$ changes $f(x)$.

$$\Delta_i = f(x) - f(x^{\text{del}(i)})$$

Here $x^{\text{del}(i)}$ denotes x with feature i replaced by the chosen baseline. The replacement rule is part of the metric.

Faithfulness correlation

Perspective:  **Content**

Property: Correctness /
faithfulness

Question: Does attribution mass correlate
with output change?

$$\text{FC}(x) = \text{corr}_S \left(\sum_{i \in S} a_i, f(x) - f(x^{\text{del}(S)}) \right)$$

Positive association is evidence that attribution mass tracks output change under the sampled subsets and replacement protocol.

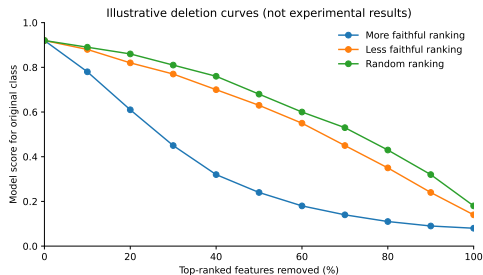
Deletion curve

Perspective:  Content

Property: Correctness /
faithfulness

Question: What happens when top-ranked
features are removed first?

Rank features by $|a_i|$, then progressively replace the most important ones. A steep score decrease early in the curve is consistent with a useful ranking under this deletion rule.



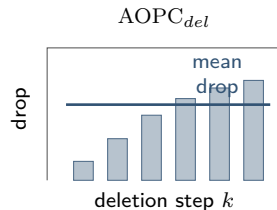
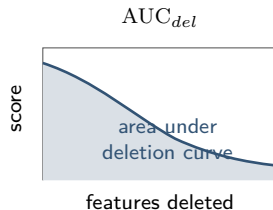
Deletion AUC and AOPC

Perspective:  **Content**

Property: Correctness /
faithfulness

Question: How do we summarize a deletion trajectory?

$$\text{AUC}_{del} = \int_0^1 f(x^{\text{del}(\alpha)}) d\alpha \quad \text{AOPC}_{del} = \frac{1}{K} \sum_{k=1}^K [f(x) - f(x^{\text{del}(k)})]$$



For a score-preserving output and a fixed deletion rule, lower deletion AUC or higher AOPC is generally preferred. Always state direction and target output.

Comprehensiveness and sufficiency

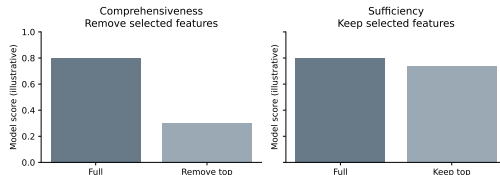
Perspective:  **Content**

Property: Completeness

Question: Is the selected evidence enough, and is removing it consequential?

$$\text{Comp}(x, S) = f(x) - f(x^{\text{del}(S)}) \quad \text{Suff}(x, S) = |f(x) - f(x^{\text{keep}(S)})|$$

- Remove selected features: does the score change? Higher drop is generally preferred.
- Keep selected features: does the score remain? Smaller difference is generally preferred.



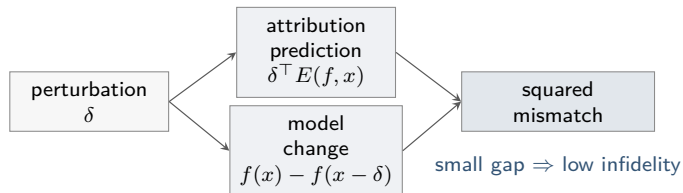
Infidelity

Perspective:  **Content**

Property: Correctness /
faithfulness

Question: Does the attribution-predicted
change match the real change?

$$\text{Infid}(E, f, x) = \mathbb{E}_{\delta} \left[\left(\delta^{\top} E(f, x) - (f(x) - f(x - \delta)) \right)^2 \right]$$



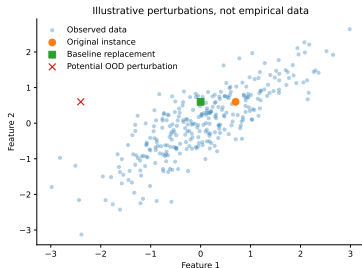
It compares attribution-predicted change with actual model change. The perturbation distribution over δ defines what changes are tested.[4]

The perturbation problem

Perspective:  Content

Property: Measurement validity

Question: Does the perturbation still represent a meaningful input?



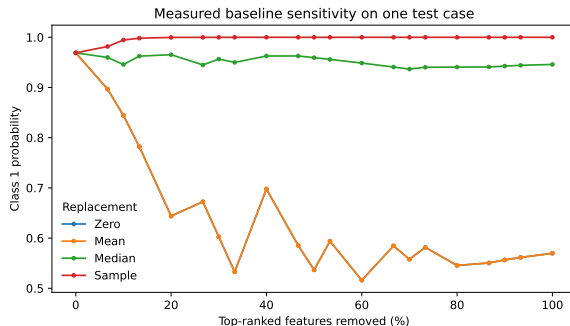
- What does “feature absence” mean: zero, mean, median, or conditional sample?
- If replacement leaves the data manifold, are we measuring the explanation or model extrapolation?

Baseline sensitivity is a result

Perspective:  **Content**

Property: Measurement validity

Question: How much does the score depend on the replacement convention?



Measured on one held-out instance; the curve changes when the replacement convention changes.

Consistency is not continuity

Perspective:  **Content**

Property: Consistency /
continuity

Question: Are we testing repeatability or
local stability?

Consistency Equivalent input/model conditions yield equivalent explanations.

Continuity Similar inputs with similar predictions yield similar explanations.

Repeated seeds and equivalent implementations probe consistency; nearby inputs probe continuity.

Max-sensitivity

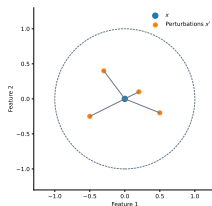
Perspective:  Content

Property: Continuity

Question: How much can the explanation change in a small neighborhood?

$$\text{Sens}_{max} = \max_{\|x' - x\| \leq r} \|E(f, x) - E(f, x')\|$$

The radius, input norm, explanation distance, and validity of perturbed inputs all matter.



Compactness is not correctness

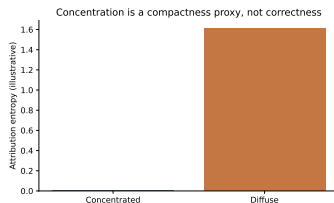
A:  **Presentation:**
Compactness

B:  **Content:** Correctness /
faithfulness

Relation: A concise explanation is easier to inspect, not automatically more faithful.

$$a = [0.91, 0.04, 0.03, 0.01, 0.01] \quad b = [0.21, 0.20, 0.20, 0.20, 0.19]$$

Which is easier to inspect? Does that make it more faithful?



Entropy and sparsity

Perspective:

 **Presentation**

Property: Compactness

Question: How concentrated is the attribution representation?

Normalize $p_i = |a_i| / \sum_j |a_j|$ and compute

$$H(a) = - \sum_i p_i \log p_i.$$

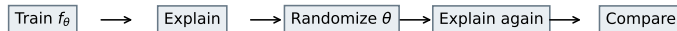
Lower entropy indicates concentration; higher Gini indicates concentration. Neither establishes that the explanation is correct or useful.

Sanity checks

Perspective:  **Content**

Property: Correctness /
faithfulness

Question: Can the protocol reject
explanations that should fail?



If explanations barely change after model parameter randomization, model dependence is questionable.

Passing a sanity check is necessary evidence, not proof of correctness.[1]

Faithful does not imply understandable

A:  **Content:** Correctness / faithfulness

B:  **Presentation:** Compactness / composition

Relation: Model-relative validity and human readability are different claims.

Faithful $\nRightarrow$ Understandable

A technically faithful explanation with a large number of predicates may be unusable for its audience.

Plausible does not imply faithful

A:  **User:** Coherence

B:  **Content:** Correctness /
faithfulness

Relation: Human plausibility is not
evidence that the explanation tracks
the model.

Coherent with human knowledge $\nrightarrow$ Correct relative to model

A domain expert may find an explanation plausible even if the model relies on a spurious feature.

Human-grounded tasks

Perspective:  User

Property: Coherence / context /
controllability

Question: Does the explanation improve a
controlled human task?

- Predict the model output; detect a model failure.
- Identify influential variables; answer comprehension questions.
- Measure decision time, confidence, and calibration.

Preference alone is weak evidence: “Which explanation do you like?” does not test task performance.